

Simplifying Surds

AQA · KS4 Year 10–11

Answer each question in your workbook. Show your working.

COMMON MISTAKES

- Not using the LARGEST square factor — do it in one step, not several.
- $(a + b) \neq a + b$. The rule only works for multiplication.
- $3 \times 12 = 3 \times 2 \times 3 = 6 \times 3$, not 3×12 .
- $a \times a = a^2$, not (a^2) left unsimplified.

METHOD

- 1 Find the largest square factor of the number
- 2 Split: $n = (\text{square} \times \text{rest})$
- 3 Use $(a \times b) = a \times b$
- 4 Square-root the perfect square out
- 5 Answer: $k \times m$ (m has no square factors)

WORKED EXAMPLE

- Simplify $\sqrt{72}$.
- Step 1 — Largest square factor: 36
- Step 2 — $\sqrt{72} = \sqrt{(36 \times 2)} = \sqrt{36} \times \sqrt{2} = 6 \sqrt{2}$

VISUAL AID

Factor tree: $72 = 36 \times 2$ $36 = 6 \times 6$ answer $6 \sqrt{2}$. Common square numbers reference: 4, 9, 16, 25, 36, 49, 64, 81, 100, 121, 144.

1. (2 marks)

Simplify: a) 12 b) 18 c) 20 d) 50

2. (2 marks)

Simplify: a) 8 b) 32 c) 45 d) 75

3. (3 marks)

Simplify: a) 98 b) 200 c) 128 d) 300

4. (3 marks)

Simplify: a) (4×7) b) 3 c) 2 d) 5 e) 8

5. (3 marks)

Simplify fully: a) $48 + 12$ b) $75 - 27$ c) $2 \times 18 + 3 \times 2$

6. (4 marks)

Show: a) $80 = 4 \times 5$ b) $3 \times 28 = 6 \times 7$ c) $180 = 6 \times 5$

7. (4 marks)

Square area = 72 cm^2 . a) Exact side? b) Perimeter? c) Compare with square of side 3 e) 2.

8. (5 marks)

a) Show $(a^2b) = a \times b$. b) Simplify $(50x^4y^2)$. c) Right triangle sides 18, 32 — exact hypotenuse?
